INCH-POUND

MIL-DTL-1608J
3 July, 2001
SUPERSEDING
MIL-H-1608H
25 November, 1991

DETAILED SPECIFICATION

HOOKS, TENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers steel wire tent hooks.

1.2 Classification. (see 6.2)

Type I – Triangle eye (hook in line with center of eye).
Size 1 - 1 inch
Size 2 - 1 ½ inch

Type III - Square eye – size 1 ½ inch
Type IV - Trapezoid eye – size 1 ½ inch
Type V - Diamond eye – size 1 ½ inch
Type VI - Diamond eye, shackle type – size 1 ½ inch

Beneficial comments (recommendations, additions, deletions, clarifications) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center Philadelphia, Clothing and Textiles Directorate, Attn: DSCP-CNR, 700 Robbins Ave., Philadelphia, PA 19111-5096 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8340

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2. APPLICABLE DOCUMENTS

*2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

TT-E-485 - Enamel, Semigloss, Rust Inhibiting

STANDARDS

FEDERAL

FED-STD-595 - Colors

(Unless otherwise indicated. copies of federal and military specifications, standards, and handbooks are available from the Defense Automated Printing Service, Bldg. 4D (DPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

*2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY BIOLOGICAL & CHEMICAL COMMAND

5-4-166 - Hooks, Tent, Assembly and Details
5-4-355 - Hooks, Tent

(Copies of drawings are available from the U. S. Army Biological and Chemical Command, ATTN: SSCNC-EMSS, NATICK, MA 01760-5017.)

*2.3 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-A-153	- Zinc Coating (Hot Dip) on Iron and Steel Hardware
ASTM-A-575	- Merchant Quality Hot Rolled, Carbon Steel Bars
ASTM-A-853	- Steel Wire, Carbon, for General Use
ASTM-B-633	- Electrodeposited Coatings of Zinc on Iron and Steel
ASTM-D-1005	- Measurement of Dry Film Thickness of Organic Coatings
ASTM-D-2092	- Preparation of Zinc-Coated Steel Surfaces for Painting

(Applications for copies of referenced documents should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19426-2959.)

*2.4 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

*3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.2 and 6.2).

*3.2 Materials and components. Materials and components shall be as specified herein and on the applicable drawings. Materials and components not definitely specified shall be of the quality normally used by the manufacturer provided the completed item complies with all the provisions of this document.

3.2.1 Wire, steel, carbon. Carbon steel wire shall conform to steel number 1010-1020, bare finish, annealed in process, of ASTM-A-853.

3.2.2 Steel bar, carbon. Carbon steel wire shall conform to grade M 1010-M 1020 of ASTM-A-575.

*3.2.3 Enamel. Enamel shall conform to type I or II, composition G or L, as applicable, olive drab color No. 24087 of TT-E-485.

3.3 Construction. Construction of the tent hooks shall conform to Drawings 5-4-166 and 5-4-355 and as specified herein.

3.3.1 Fabrication.

3.3.1.1 Type I, III, IV and V tent hooks. The body for the type I, III, IV and V tent hooks shall be fabricated from steel wire specified in 3.2.1. The ends of the formed wire body shall be securely welded as shown on Drawing 5-4-355 and all welds shall be complete, sound and uniform. The type I, size 1 tent hook shall be capable of withstanding a load of 125 pounds minimum in tension without any evidence of cracked or fractured weld or permanent distortion of the wire body when tested as specified in 4.4.5. The type I, size 2 and types III, IV and V tent hooks shall be capable of withstanding a load of 500 pounds minimum in tension without any evidence of cracked or fractured weld or permanent distortion of the wire body when tested as specified in 4.4.5.

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3.3.1.2 Type VI tent hook. The hook portion shall be fabricated of steel wire specified in 3.2.1. The upper and lower shackles shall be fabricated of steel bar specified in 3.2.2. The welding of the hook to the upper shackle shall be complete, sound and uniform. After enameling, the tent hook shall be assembled with bolt and the self locking nut shall be drawn up finger-tight.

3.3.2 Deburring. All burrs, sharp edges, and rough spots shall be removed prior to application of the finish.

3.4 Finish.

3.4.1 Zinc coating. After complete fabrication of the tent hooks and prior to assembly of the type VI tent hook, all tent hooks and components (except bolt and nut) shall be zinc coated. The zinc coating shall conform to either ASTM A 153 with preparation for painting in accordance with method A, crystalline zinc phosphate conversion treatment of ASTM D 2092 or to type IV, thickness classification Fe/Zn12 of ASTM B 633. As an alternate, use chromate conversion type II of ASTM B 633.

3.4.2 Enameling. All zinc coated surfaces of tent hooks and components (except bolt and nut) shall be completely coated with enamel specified in 3.2.3. The coating shall be a smooth, uniform, dry film without runs, blisters, wrinkles, area of thin or no film, scratch or scuff, foreign matter imbedded in finish, and wet or tacky paint.

3.4.3 Epoxy powder coating. As an alternate to the zinc coating and enameling specified in 3.4.1 and 3.4.2 the surfaces shall be prepared and coated as follows:

a. The tent hooks shall be prepared for coating by mechanical means. The surfaces shall be thoroughly cleaned and uniformly textured by blasting using 80 grit aluminum oxide. Prepared surfaces shall be kept clean during subsequent handling. Handling shall be conducted by clean gloved hands or by clean implements. The surfaces shall be epoxy coated within 8 hours after sand blasting.

b. The finished coating shall be camouflage green 483 conforming to color chip 34094 of FED-STD-595 and finished with an electrodeposited epoxy powder coating. The epoxy coating shall conform to epoxy coatings produced by H. B. Fuller Co., or the O'Brien corporation, or equal (see 6.6 and 6.8).

c. The electrodeposited epoxy powder coating shall be baked on in accordance with commercial practices. The thickness of the coating shall be .003 $\pm$ .001 inch.

d. The final coating shall be smooth and uniform in appearance and shall be free from wrinkles, streaks, corrosion, blistering, peels, cracks or sags.

3.5 Marking for identification. The hooks shall be marked with the manufacturer's identification in accordance with industry practice.

3.6 Workmanship. All welding shall be free from , pits, slag inclusion, porosity, crack, flash and excess flux deposit. The finished tent hooks shall be clean and free of dirt and grease. Threads and finish of bolt and nut shall be free from damage, i.e., nut shall not bind on bolt. The tent hooks shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

*3.7 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. When a first article is required (see 6.2) the tent hooks it shall be examined for the defects listed in 4.3.4 and 4.3.5 and tested for the strength characteristic specified in 4.6.1.

4.3 Inspection.

4.3.1 Conformance inspection. Sampling for inspection shall be as specified in the contract or purchase order.

4.3.2 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.3 In-process inspection. Inspection shall be made of the operations and requirements listed in Table I to establish conformance with the specified requirements. Whenever nonconformance is noted, correction shall be made to the items affected and the operations. Items which cannot be corrected shall be removed from production. The government reserves the right to exclude from consideration for acceptance any material for which in-process inspection has indicated nonconformance.

Table I. In-process inspection.

Requirement or operation	Requirement paragraph
Removal of burrs, sharp edges and rough spots prior to finishing	3.3.2
Zinc coating of tent hooks and components (except bolt and nut) prior to enameling	3.4.1

*4.3.4 End item visual examination. The end item shall be examined for the defects listed in Table II. The lot size shall be expressed in units of frame sections of one type only. The sample unit shall be one completely fabricated frame section. The inspection level shall be II.

TABLE II. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Color of finish is not as specified		201
	Not finished	101	
	Coating not smooth or uniform		202
	Finish is not dry, i.e., wet or tacky to touch	102	
	Run, wrinkle or blister		203
	Area of no film exposing bare metal	103	

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TABLE II. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Area of thin film exposing zinc coating		204
	Any scratch or foreign matter embedded in finish		205
Construction & workmanship	Any component not as specified or missing, e.g., shackle, bolt or nut	104	
	Type IV tent hook not assembled or the self-locking nut has not been drawn up finger-tight		206
	Threads or finish of bolt or nut is damaged, i.e., nut binds on bolt		207
	Not clean, i.e., has dirt or grease		208
	Burr, sharp edge, or rough spot		209
Welding	Missing where required	105	
	Not type specified	106	
	Incomplete or not sound	107	
	Not uniform		210
	Pits, slag inclusion or porous		211
	Crack or fracture	108	
	Flash or excess flux deposit has not Been removed		212
Marking for identification	Missing, illegible, incorrect or incomplete		213

*4.3.5 End item dimensional examination. Examination shall be made of the end item for compliance with the dimensions on the drawings. Only those dimensions that can be evaluated without damaging or disassembling the end items shall be examined. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of tent hooks of one type only. The sample unit shall be one tent hook. The inspection level shall be S-2.

*4.3.5.1 Epoxy coating thickness examination (when applicable). The end item shall be examined for thickness of the epoxy coating in accordance with ASTM-D-1005. Three observations shall be made per sample unit. Any epoxy coating with a minimum thickness not as specified in 3.4.3 shall be classified as a defect. The lot size shall be expressed in units of tent hooks of one type offered for inspection at one time. The sample unit shall be one tent hook. The inspection level shall be S-2.

*4.4 End item testing. Types I, III, IV and V tent hooks shall be tested for strength as specified in 4.6.1. The inspection level shall be S-2. The requirement shall be applicable to the individual unit. There shall be one determination per sample unit for types I, III and IV hooks and two determinations per sample unit for type V hooks.

*4.5 Packaging examination. The fully packaged end items shall be examined for marking, materials, workmanship and content defects.

4.6 Methods of inspection

4.6.1 Strength test. The tent hooks shall be tested as shown on Drawing 5-4-355. The load rate shall not exceed 1000 pounds per minute. The type I, size 1 tent hook shall be subjected to a pull of 125 pounds minimum. The type I, size 2, III, IV and V tent hooks shall be subjected to a pull of 500

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pounds minimum. After the test load has been released, the tent hooks shall be examined. Any tent hook having evidence of cracked or fractured weld or permanent distortion of the wire body shall be classified as a defect.

5. PACKAGING

*5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The tent hooks are intended for use on various types of tents and shelters.

*6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Type and size of tent hook required (see 1.2).
c. When first article is required (see 3.1, 4.2 and 6.4).
d. Levels of preservation and packing (see 5.1).

6.3 Sample. For access to standard sample, address the contracting activity issuing the invitation for bids.

6.4 First article. When a first article is required, it will be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

*6.6 Manufacturers of epoxy powder paint products. Epoxy paint conforming to the requirements of 3.6.4 may be obtained from the following manufacturers:

H.B. Fuller Co.
North American Adhesives, Sealants
& Coatings Group
3530 Lexington Ave. N.
St. Paul, Minnesota 55110

O'Brien Powder Products, Inc.
9800-T Genard Rd.
Houston, TX 77021

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6.7 International standardization agreements. Certain provisions of this document are the subject of international standardization agreements as cited in NATO, STANAG NO. 2882, relative to camouflage requirements for tents, shelters and subsidiary components. When amendment, revision or cancellation of this document is proposed that will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accommodations.

*6.8 Subject term (key word) listing.
Attachment fixtures
Fasteners

*6.9 Equal item. Prior to the use of an "or equal" item, the supplier must submit the item, with supporting data to the contracting officer for subsequent approval or disapproval by the responsible military agency.

6.10 Changes from previous issue. The margins of this document have been marked with an asterisk(*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, as written, irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - GL
Navy – NU
Air Force – 11.6

Preparing activity
DLA-CT

Project No: 8340-0591

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7 and send to preparing activity.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-DTL-1608J	**2. DOCUMENT DATE** ***(YYYYMMDD)*** 20010703

3. DOCUMENT TITLE
Hooks, Tent

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) DSN *(If applicable)*	7. DATE SUBMITTED *(YYYYMMDD)*

8. PREPARING ACTIVITY

a. NAME DEFENSE SUPPLY CENTER PHILADELPHIA DSCP-CNRP	b. TELEPHONE *(Include Area Code)* (1) Commercial (215) 737-3290 (2) DSN 444-3290
c. ADDRESS *(Include Zip Code)* 700 Robbins Ave (Bldg 6, C&T) PHILADELPHIA, PA 19111-5092	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort Belvoir, Virginia 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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